

Wooseok Song

B.S. Candidate in Artificial Intelligence

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Education

Gachon University

B.S. in Artificial Intelligence, GPA: 4.09/4.5

Seongnam, Republic of Korea

Expected Feb. 2027

Research Interests

- Learning-based dense SLAM and 3D scene priors
- Robust SLAM in dynamic and degraded environments

Publication

W. Song, J. Cho, and J. Ahn, “Enhancing Lightweight End-to-End Autonomous Driving via Data Augmentation,” 2025 KSAE Annual Autumn Conference and Exhibition, 2025. [Domestic, in Korean]

Research Experience

RAISE Lab, Gachon University

Undergraduate Research Intern

Seongnam, Republic of Korea

Apr. 2026 – Present

- Reviewed recent work on learning-based dense SLAM and feed-forward 3D reconstruction.
- Ran the public SLAM-Former implementation on EuRoC, KITTI, and indoor image sequences, and documented results and failures in low-texture, repetitive, and long-sequence settings.

P-Career Catch Research Program, Gachon University

Undergraduate Research Project - 3D LiDAR Scan-Map Change Detection

Seongnam, Republic of Korea

Feb. 2026 – Jul. 2026

- Investigated how visibility and occlusion affect insertion and removal labeling in 3D LiDAR scan-map change detection.
- Generated controlled reference-map and current-scan pairs in CARLA and Isaac Sim under structural changes and different LiDAR configurations.

Selected Projects

Parang-E Control - 3D LiDAR SLAM

VIP-LAB, Gachon University

Aug. 2025 – Dec. 2025

- Tested LeGO-LOAM, KISS-SLAM, and FAST-LIO-SAM systems using MORAI, CARLA, and collected driving data.
- Integrated the resulting point-cloud maps with hdl_localization and debugged ROS topics, frames, and sensor configurations.

OpenHouse - Smartphone-Based Indoor 3D Reconstruction

Capstone Design Project, Gachon University

Mar. 2026 – Jul. 2026

- Built an indoor 3D reconstruction pipeline from smartphone images to a mobile-viewable 3D model.
- Tested COLMAP, DUST3R, VGGT, and MapAnything on indoor image sequences and selected a pipeline based on qualitative results and execution constraints.
- Implemented ceiling removal and tuned Poisson surface reconstruction for mesh generation.

LIMO Autonomous Driving Competition

VIP-LAB, Gachon University

Jul. 2025 – Aug. 2025

- Developed a camera- and 2D LiDAR-based pipeline for lane following and obstacle avoidance.
- Tested rule-based lane detection under changes in illumination and background.
- Integrated and debugged the complete driving pipeline.

Awards & Honors

1st Place, LIMO Autonomous Driving Competition, Gachon University

Aug. 2025

Excellence Award, Capstone Design Poster, Department of Artificial Intelligence, Gachon University

Jun. 2026

Activities

Take Out Autonomous Driving Club

Gachon University

Sep. 2024 – Dec. 2025

- Participated in the AME Autonomous Driving Hackathon (Jul. 2025).
- Participated in the Autonomous Driving AI Challenge (Aug. – Sep. 2025).
- Mentored a ROS and OpenCV study (Nov. 2025).

Technical Skills

Programming: Python, C++, Bash

Simulation: NVIDIA Isaac Sim, CARLA, MORAI

Language: TOEIC Speaking - IH (Mar. 2026)

Robotics: ROS, ROS 2

Development: Ubuntu, Docker, Git